Article Review

The motivation of senior clinicians to teach medical students

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What motivates senior clinicians to teach medical students?

A research question for this study is to assess the motivation of senior medical clinicians to teach medical students. Although this is a simple research question, it clearly describes the relationship between senior medical clinicians and the motivation to teach medical students. This study is a form of applied research that seeks ways to improve the recruitment and retention of important clinical teachers. In addition, there are adequate resources available to conduct this type of study, such as sampling of the population, research variables, instruments, methods, and data collection tools (pg. 3). Furthermore, this study is aimed at identifying the factors in the design of teaching programs that meet teacher motivations and ultimately enhance the effectiveness of the medical teaching workforce. Motivation is a key factor in determining the drive behind clinicians' behavior to teach medical students. The statement rankings are measuring instruments to determine the clinician's motivation (see Figure 1). The data and analysis from this study clearly show a correlation between intrinsic factors such as altruism, intellectual satisfaction, personal skills, and truth-seeking and the motivation to teach students. There is also strong evidence that reasons for not teaching included a lack of strong involvement in course design, a heavy clinical load, or a feeling that it was a waste of time (pg. 5).

Methods

The data collection method is Q-sort. The population chosen for the Q sample is drawn from various clinician specialties, ages, and genders, and the sample is drawn exclusively from a public, tertiary-level regional teaching hospital. Subjects were 75 participants of the study group of 101 senior medical clinicians registered on the medical student teaching staff. Demographic data collected included gender, age group (30-40, 41-55, >56 years), employment contract (salaried staff specialist or contacting visiting medical officer), and the clinician's specialty (internist, surgeon, or other sub-specialist—pathologists, radiologists, psychiatrists, etc.).

Of the 75 participants, 18 (24%) were female, and 57 (76%) were male, with the following age distributions: 30-40 = 16 (21.3%), 41-45 = 46 (61.3%), and greater than 55 = 13

(17%). Most were staff specialists (n = 48, 64%), and 27 (36%) were visiting medical officers. Half of the participants were internists (n = 39, 52%), 12 (16%) were surgeons, and 24 (32%) were other groups (pg. 5).

Participants' motivations for teaching medical students were assessed using Q methodology. Q methodology combines qualitative and quantitative research and uses standardized instruments, statistical analyses, and face-to-face interactions. This method allows for a quantitative evaluation of individuals' opinions on topics of common concern (pg. 3).

Overall, Q methodology is a good method for this kind of study, which uses rankings of individual statements on a continuum to indicate the degree of agreement or disagreement with each statement. This method ranks individual opinions on the statements, facilitating the formation of individual viewpoints on the overall subjective questions to which the sample statements referred (see Figure 1). The viewpoints are then compared in a correlation matrix to identify similarities between individual viewpoints.

Instrument

The data collection instrument is a table (11 columns by 11 rows) that contains a series of cells arranged in a symmetric, forced normal distribution (see Figure 1). Each participant is given a stack of 69 note cards containing questions and is instructed to sort all cards by comparing each statement to the others (pg. 3). The Q-sort instruction is simple but a bit confusing because each participant has to sort and prioritize all the note cards. The ranking is an individual statement comparison to the other statements. The final positioning of all statements, therefore, provides a ranking of their relative importance as determined by the individual clinician. A simpler approach is to develop satisfaction ratings, such as a 5-point scale (strongly disagree, disagree, neutral, agree, strongly agree), and collect open-ended comments to capture the qualitative aspects of Q methodology.

Variables

The variables for analysis are the statement rankings produced by the study group of clinicians who share the same viewpoints as other clinicians. There were 69 statements (the variables), and approximately equal numbers of positive and negative statements were created (see Table 1). Positive scores indicated agreement, while negative scores indicated disagreement. Decreasing scores reflected less important views. The statements chosen for the Q sample are well constructed. It may be a good idea to conduct a study of each individual statement, which may tell the researcher more about specific issues. Nevertheless, ranking these statements in this study can help researchers identify the most important issues and filter out the less important ones.

Other variables in this study include gender, age group (30-40, 41-55, > 56 years), employment contract (salaried staff specialists or contracting visiting medical officer), the specialty of the clinician—internist, surgeon, or other sub-specialists—pathologists, radiologists, psychiatrists, etc. (pg. 2).

Statistical Test

The Q sample is sorted into four factors: Factor 1 (Table 2, accounting for 68 participants' sorts), Factor 2 (Table 3, 3 sorts), Factor 3 (Table 4, 2 sorts), and Factor 4 (Table 5, 2 sorts). To Factor 1, the statements phrased beginning with "I teach because…", most participants agree with positive statements about the value of teaching and disagree with statements phrased negatively about teaching (pg. 5). In Factor 2, 3, and 4, the statements phrased with "I struggle to teach because…" are less favorable. Based on Maslow's human needs theory (1954, which motivates behavior—physiological needs, safety needs, belonging needs, esteem needs, the need for self-actualization—is the drive in ranking the Q sample (or statements). There is strong evidence that all of these needs influence the ranking of the statements in this study.

Based on the data of this study, the highly ranked statements include: 1) helping students become good doctors, 2) enjoying the challenge of effective teaching, 3) valuing the representation of one's own specialty, 4) enjoyment of small group teaching, 5) inspiration from mentors and past teachers, 6) liking to be challenged in one's views, 7) feeling responsible for students, and 8) wanting to understand student. The highly ranked negative statements include: 1) lack of involvement in course design, 2) lack of enjoyment in teaching, and 3) clinical load deterring involvement in teaching (pg. 7-8).

Conclusion

This study has successfully determined the motivation of senior clinicians to teach medical students. Further, I think that data from this study can be used to enhance the effectiveness of the medical teaching workforce. The data suggest a correlation between medical clinicians' behavior and motivation to teach. Furthermore, I think the data can be used to improve the recruitment and retention of clinician-teachers. However, I feel that Q sorting may not be a good method for this study because there are just too many factors, such as the Q sample (69 statements), sorting the Q note cards, evaluating the questionnaires, and physically placing all the questions in the Q sort. Overall, the data collection method is just too complicated and time-consuming.

Finally, this study promotion of teaching to senior clinicians is likely to have increased success if prospective teachers contribute to course development, sufficient time is allocated to teaching, memories of inspirational teachers are reawakened, the link between strong teaching and junior doctor outcomes is emphasized, and staffs are reminded of the opportunity to advertise their specialty (pg. 9). I think that is continuing to be a challenge to motivate senior medical clinicians, because rewards and incentives for teaching just do not exist. I think that drive and motivation are purely based on individual clinician behavior. I hate to say this, but further study is needed to determine the relationship between the motivation to teach and gender, age group, employment contract, and other specialties.

References

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